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Survey paper

A survey on 6G and O-RAN intelligence: Semantic protocols, protocol learning, and AI-enabled semantic protocols

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ABSTRACT

This paper presents a comprehensive survey of semantic protocols, protocol learning, and AI-enabled semantic protocols within the context of Open RAN and 6G networks. We systematically review the significant progress achieved in these domains, highlighting key methods such as transformer-based semantic encoders, reinforcement learning-driven protocol adaptation, and federated learning frameworks for distributed training. Across surveyed studies, notable achievements include bandwidth savings of 35–70%, improved robustness under noisy conditions, and enhanced interoperability in multi-vendor environments. By consolidating findings, we identify major challenges such as the lack of standardized semantic KPIs, computational overhead at the edge, interoperability issues, and emerging security vulnerabilities. Furthermore, we categorize open research opportunities into theoretical, methodological, technical, and implementation directions, providing a clear roadmap for future development. This survey ultimately positions semantic communication and AI-enabled protocols as pivotal enablers for meaning-centric, adaptive, and efficient next-generation O-RAN/6G networks.

1. Introduction

1.1. Motivation

The telecommunications landscape is undergoing a fundamental transformation, evolving from basic data-transmission systems to intelligent frameworks supporting next-generation applications. Contemporary networks must accommodate unprecedented requirements for extended reality (XR), tactile internet services requiring ultra-low latency, immersive holographic communications, and Industry 5.0's advanced connectivity demands. These applications require networks that comprehend, process, and prioritize information based on semantic meaning rather than merely providing higher bandwidth [1]. Extended reality applications demand transmission of contextual information preserving spatial relationships beyond simple image data. The tactile internet requires networks understanding the criticality of sensory feedback loops for real-time actuation. Holographic communications need networks prioritizing image components most crucial for three-dimensional representation. Industry 5.0's human-machine collaboration necessitates networks recognizing semantic importance across diverse data streams in manufacturing environments [2,3]. The traditional Shannon-inspired communication paradigm, focused on bit-accurate transmission irrespective of content meaning,treats all data equally an approach proving inefficient when meaning and context are paramount. This limitation becomes critical under bandwidth restrictions where perfect data reproduction is impractical. Open Radio Access Networks (O-RAN) and sixth-generation (6G) wireless systems offer architectural paradigms conducive to semantic communication. O-RAN's disaggregated, multi-vendor ecosystem provides flexibility while requiring intelligent coordination. Its programmability and open interfaces create favorable conditions for semantically aware protocols that adapt dynamically. Similarly, 6G networks are envisioned as intelligent systems interpreting context and information importance. Semantic protocols offer a paradigm shift by reorienting communications toward meaning preservation rather than perfect bit reproduction. By focusing on semantic content—what data means rather than its representation—these protocols enhance efficiency, dependability, and user experience. Combined with protocol learning and AI methods, they enable networks to extract

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communication essentials, allocate resources based on semantic weighting, and make contextually informed decisions, transforming networks from passive conduits into active participants that intelligently process, learn from, and reason about information [2,3].

1.2. Objective

This paper conducts a comprehensive survey of semantic protocols, protocol learning, and AI-enabled semantic approaches, focusing on their applications in O-RAN and 6G architectures. Our primary objective is synthesizing existing work to provide a conceptual roadmap highlighting implications of semantic-aware approaches for researchers, network architects, and telecommunications experts. We evaluate theoretical foundations, current implementations, and future prospects of semantically enhanced networks. Specifically, we analyze how semantic protocols enhance flexibility and efficiency by emphasizing meaning over perfect transmission; how protocol learning enables optimization through experience; and how AI supports semantic understanding in network protocols. These methods are positioned within O-RAN and 6G paradigms, defining opportunities and constraints for meaning-centric communication. This review synthesizes theoretical advancements with practical considerations, identifies research gaps, and suggests future directions [4,5].

1.3. Scope and contributions

This survey makes several contributions to the field of semantic-aware networking within O-RAN and 6G architectures. First, it provides a comprehensive survey of foundational concepts underlying semantic communication, protocol learning, and AI-enabled semantic protocols, tracing their evolution from theoretical origins to current research, clarifying terminology, and organizing the literature into a coherent structure [6,7]. Second, it introduces a high-level conceptual framework that synthesizes insights from existing works into an integrated O-RAN/6G architecture. This framework is presented as a unifying synthesis that highlights connections across prior studies and identifies promising research directions rather than as an implementation-ready proposal. Third, the survey systematically identifies technical challenges, theoretical gaps, and deployment issues facing semantic-aware networks, including limitations in semantic reasoning frameworks, scalability concerns, interoperability challenges, and computational complexity issues. Fourth, it proposes focused avenues for future research that span theoretical developments, algorithmic innovations, and architectural enhancements, aiming to stimulate targeted efforts that advance the field. Finally, it outlines a conceptual roadmap that bridges theoretical models with real-world deployment considerations, addressing standardization needs and performance evaluation frameworks. By explicitly framing this work as a survey and conceptual synthesis, the paper seeks to accelerate the development and adoption of semantic-aware networking approaches within O-RAN and 6G contexts, providing both a reference for researchers and a guide for practitioners [8,9]. Fig. 1 presents a high-level conceptual framework synthesized from prior works integrating Semantic Protocols, Protocol Learning, and AI-enabled communication into O-RAN and 6G architectures. User Equipment and IoT devices communicate through semantic encoders, supported by Edge Devices with context-awareness engines. These feed into the RAN Intelligent Controller (RIC), managing radio resources using AI and Protocol Learning models via xApps and rApps. A Semantic Layer spans the architecture, interacting with shared Knowledge Base and Federated Learning Systems ensuring decentralized, privacy-preserving training. The architecture incorporates Reconfigurable Intelligent Surfaces and Terahertz links, emphasizing scalable, intelligent, context-aware communication.

Fig. 2 depicts the proposed five-layer architecture for O-RAN/6G networks synthesized from existing studies, integrating semantic protocols, protocol learning, and AI-enabled intelligence to transform bit-centric systems into context-aware, self-optimizing platforms. The Service Layer hosts demanding 6G use cases: holographic communications, digital twins, extended-reality applications, and tactile-internet services. The Service Management and Orchestration (SMO) Layer operates with Non-RT RIC training ML models, Semantic Protocol Designer leveraging centralized Knowledge Base, and Cross-Domain Orchestrator harmonizing end-to-end QoS targets. The Intelligent Control Layer bridges planning and execution through Near-RT RIC's Real-Time Semantic Analyzer, semantic xApps, Protocol Learning Agents, and Semantic Intelligence Plane. The Network Functions Layer transforms RAN components into AI-Enabled Semantic RAN (S-RAN) with intelligent processing units. The Infrastructure Layer provides AI Processing Units, semantic middleware, edge computing nodes, and core network with O-RAN interfaces. Solid arrows indicate data and semantic information flow, while dashed arrows represent control signals and feedback loops. Knowledge flows from Knowledge Base into Semantic Intelligence Plane; control directives propagate from RIC to semantic RAN elements; federated learning updates traverse between edge devices and controllers. These loops enable continuous network evolution learning from performance, adapting protocols to contexts, and dynamically allocating resources. By embedding semantic understanding, learning-based adaptation, and AI-driven control at every layer, this architecture delivers ultra-efficient spectrum utilization, sub-millisecond latency with high reliability, and substantial energy savings, transforming O-RAN and 6G deployments into cognitive entities meeting stringent requirements of intelligent applications [10,11].

Section II reviews Background and Key Concepts, covering semantic communication, protocol learning, and AI-enabled semantic protocols. Section III surveys Related Work, detailing existing integration efforts in O-RAN and 6G. Section IV examines Application to O-RAN Architectures, illustrating semantic and AI-based protocol deployment. Section V analyzes Challenges, Research Gaps, and Future Research Directions in a unified manner, including standardization, real-time inference, security, interoperability, semantic KPIs, lightweight inference, and cross-domain orchestration. Finally, Section VI concludes with key insights and outlines the roadmap toward meaning-centric O-RAN/6G networks. A comprehensive list of all abbreviations used throughout this paper is provided in Table 1.

2. Background and key concepts

This section establishes the theoretical and technical foundations for understanding the integration of semantic protocols, protocol learning, and AI-enabled semantic approaches in next-generation networks. By tracing the evolution of communication paradigms and synthesizing cutting-edge advancements, we provide a framework for analyzing their role in O-RAN and 6G architectures. Section II-A focuses strictly on semantic protocols; Section II-B on protocol learning; and Section II-C on AI-enabled semantic protocols. Each subsection emphasizes unique aspects without overlap.

2.1. Semantic protocols

Semantic protocols represent a fundamental paradigm shift from Shannon's classical model, which prioritizes the reliable delivery of bits regardless of content meaning, to systems that emphasize the preservation and interpretation of contextual significance [10,11]. While Shannon's framework revolutionized telecommunications by formalizing concepts such as channel capacity,

$$C = B \log_2 \left(1 + \frac{S}{N}\right), \tag{1}$$

its bit-centric approach treats all data equally, ignoring the inherent meaning in the transmitted information.

The theoretical foundation of semantic protocols centers on semantic entropy, which quantifies information based on meaning rather than

Table 1
List of abbreviations.

Abbreviation	Definition
3GPP	3rd Generation Partnership Project
5G	Fifth Generation (of wireless communication systems)
6G	Sixth Generation (of wireless communication systems)
AI	Artificial Intelligence
AI-ML	Artificial Intelligence and Machine Learning (as a combined term in O-RAN specifications)
AUC	Area Under the Curve (a metric for evaluating classification models)
CNN	Convolutional Neural Network
CU	Centralized Unit (in O-RAN, handles RRC, PDCP, and higher-layer functions)
D2D	Device-to-Device (direct communication between user equipment)
DeepSC-XL	A specific transformer-based model for semantic communication
DNN	Deep Neural Network
DU	Distributed Unit (in O-RAN, handles upper PHY, MAC, and RLC layer functions)
E2	O-RAN's interface between the Near-RT RIC and the O-DU/O-CU
E2AP	E2 Application Protocol
E2SM	E2 Service Model
FRAND	Fair, Reasonable, and Non-Discriminatory (licensing terms for essential patents)
GAN	Generative Adversarial Network
GNNs	Graph Neural Networks
HARQ	Hybrid Automatic Repeat Request
IoT	Internet of Things
KPIs	Key Performance Indicators
LLM	Large Language Model
MAC	Media Access Control (a sublayer of the data link layer)
MDP	Markov Decision Process
MEC	Multi-access Edge Computing
ML	Machine Learning
Near-RT RIC	Near-Real-Time RAN Intelligent Controller
NLP	Natural Language Processing
Non-RT RIC	Non-Real-Time RAN Intelligent Controller
O1	O-RAN's management and orchestration interface
O2	O-RAN's interface for semantic traffic steering policies
O-Cloud	O-RAN Cloud infrastructure
O-CU	O-RAN Centralized Unit
O-DU	O-RAN Distributed Unit
O-RAN	Open Radio Access Network
PDCP	Packet Data Convergence Protocol
PHY	Physical Layer (Layer 1 of the OSI model)
QoE	Quality of Experience
QoS	Quality of Service
RAN	Radio Access Network
RF	Radio Frequency
RIC	RAN Intelligent Controller
RIS	Reconfigurable Intelligent Surfaces
RLC	Radio Link Control (a Layer 2 protocol in wireless stacks)
RL	Reinforcement Learning
RNN	Recurrent Neural Network
RRC	Radio Resource Control (a Layer 3 protocol for control plane management)
RU	Radio Unit (in O-RAN, handles lower PHY layer and RF functions)
rApps	non-real-time applications running on the Non-RT RIC for policy and analytics
SAC	Soft Actor-Critic (a reinforcement learning algorithm)
SHAP	SHapley Additive exPlanations (a method for explaining AI model outputs)
SMO	Service Management and Orchestration
S-RAN	Semantic RAN (a Radio Access Network enhanced with semantic-aware processing)
THz	Terahertz (referring to the high-frequency radio spectrum band for 6G)
TIFG	O-RAN Testing and Integration Focus Group
TinyML	Tiny Machine Learning (running ML models on highly resource-constrained edge devices)
TR	Technical Report (a type of document from 3GPP)
UE	User Equipment
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
WGAN-GP	Wasserstein Generative Adversarial Networks with Gradient Penalty
WG3	O-RAN Working Group 3 (responsible for the Near-RT RIC and E2 interface)
WG6	O-RAN Working Group 6 (responsible for Cloudification and Automation)
xApps	real-time applications running on the Near-RT RIC for control and optimization
XR	Extended Reality (encompassing augmented, virtual, and mixed reality)

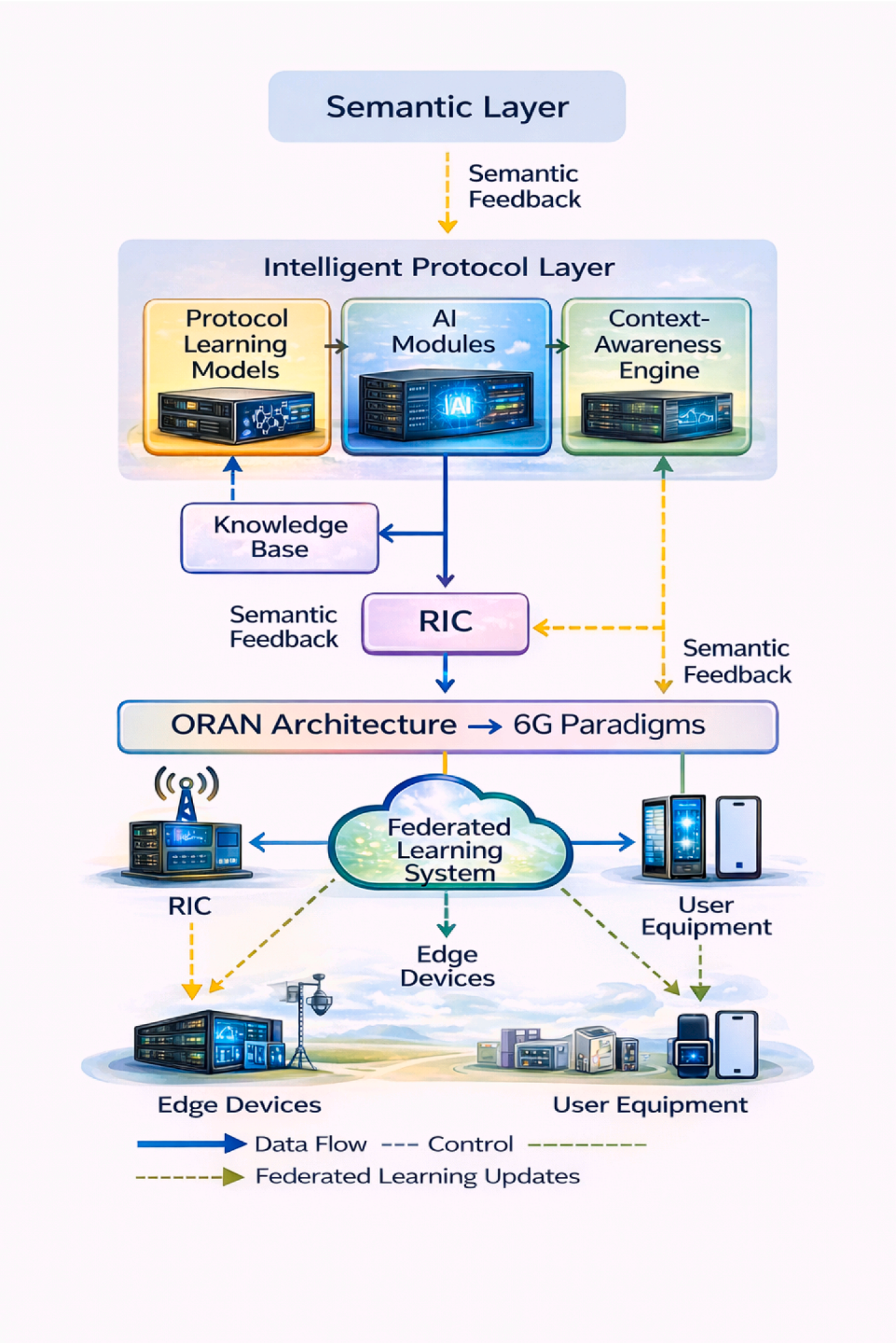


Fig. 1. Conceptual illustration of how semantic protocols, protocol learning, and AI-enabled modules have been discussed in the literature for integration into O-RAN/6G systems. This synthesis highlights common architectural elements such as semantic encoders, AI-driven RICs, federated learning, and context engines.

statistical frequency:

$$H_s(X) = -\sum_{x \in \mathcal{X}} P_s(x) \log P_s(x), \tag{2}$$

where $P_s(x)$ represents the probability distribution over semantic importance rather than raw data frequency [12,13]. This mathematical framework enables meaning-centric compression by identifying and preserving semantically critical information while discarding redundant data.

Semantic protocols achieve substantial efficiency gains through semantic-aware encoding techniques. For example, in holographic communications, semantic encoders distinguish between critical depth-perception metadata and redundant background details, achieving compression ratios exceeding 50:1 without perceptual quality loss. The semantic encoding process transforms raw data into semantic representations that preserve contextual meaning while minimizing transmission overhead.

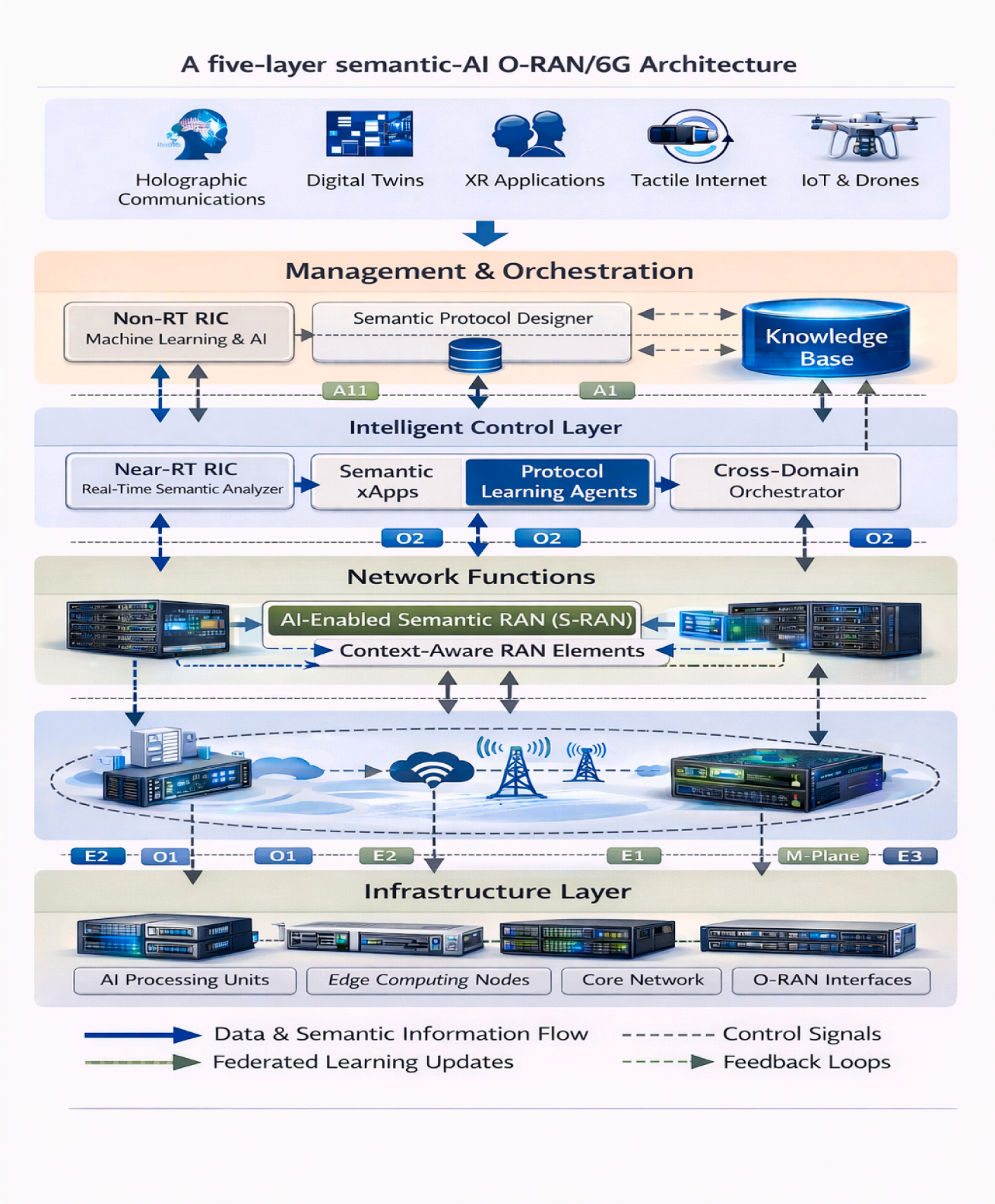


Fig. 2. A five-layer semantic-AI O-RAN/6G architecture that transforms application intents into formal KPIs, orchestrates them via AI-driven SMO/RIC modules, adapts protocols in real time with neuro-symbolic xApps, embeds context-aware processing in RAN units, and leverages federated learning and AI accelerators to create continuous feedback loops for self-optimizing, resilient networks.

The efficiency benefits of semantic protocols include significantly improved bandwidth utilization through meaning-centric data reduction, enhanced communication reliability by preserving critical semantic content, and optimized resource allocation based on semantic importance rather than data volume. These protocols fundamentally redefine communication efficiency by focusing on the successful transmission of meaning rather than bits [14,15].

2.2. Protocol learning

Protocol learning introduces dynamic optimization into communication systems through the systematic application of machine learning algorithms to network protocol adaptation. Unlike static, rule-based protocols, protocol learning enables networks to autonomously modify their operational strategies based on observed performance patterns and environmental feedback [8,9].

The mathematical foundation of protocol learning models the network as a Markov Decision Process:

$$\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle,$$

where $\mathcal{S}$ represents the state space of network conditions, $\mathcal{A}$ defines the available protocol actions, $\mathcal{P}$ captures state transition probabilities, $\mathcal{R}$ provides reward functions for performance evaluation, and γ represents the temporal discount factor for future rewards.

Supervised learning approaches in protocol learning employ graph neural networks (GNNs) to learn optimal protocol parameters from labeled training datasets. These networks excel at capturing the topological relationships inherent in network structures, enabling accurate prediction of optimal protocol configurations. Recent implementations have demonstrated prediction accuracies of approximately 92% for beamforming optimization tasks under varying signal-to-interference-plus-noise ratio conditions [16].

Reinforcement learning methods, particularly the Soft Actor-Critic (SAC) algorithm, enable protocols to learn optimal policies through trial-and-error interactions with the network environment. The continuous policy optimization allows for fine-grained adaptation to dynamic network conditions, with reported performance improvements in resource allocation efficiency and latency reduction.

Federated learning represents a distributed approach to protocol learning that enables collaborative optimization across multiple network nodes while preserving data privacy. This approach allows individual network elements to contribute to global protocol optimization without sharing sensitive operational data, achieving reported improvements in handover success rates of up to 28% [17–19].

2.3. AI-enabled semantic protocols

AI-enabled semantic protocols represent the convergence of semantic processing capabilities with advanced artificial intelligence techniques to create autonomous, context-aware communication systems. These protocols integrate sophisticated deep learning models, anomaly detection mechanisms, and adaptive routing algorithms to enable intelligent semantic processing and decision-making [20,21].

Deep learning models form the computational backbone of AI-enabled semantic protocols. Transformer-based architectures, particularly those utilizing attention mechanisms, enable sophisticated context extraction from high-dimensional semantic inputs:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \tag{3}$$

where Q, K, and V represent query, key, and value matrices respectively. Advanced implementations like DeepSC-XL leverage these attention mechanisms to achieve task-specific feature extraction, resulting in payload size reductions of up to 70% in vehicular network applications [22,23].

Anomaly detection represents a critical capability within AI-enabled semantic protocols, enabling real-time identification of semantic inconsistencies and network irregularities. Wasserstein Generative Adversarial Networks with Gradient Penalty (WGAN-GP) have demonstrated exceptional performance in this domain, achieving Area Under the Curve (AUC) scores of 99.3% for network performance monitoring tasks. These models excel at distinguishing between normal semantic patterns and anomalous behaviors that could indicate security threats or system malfunctions.

Adaptive routing algorithms within AI-enabled semantic protocols utilize graph neural networks to optimize data flow based on semantic content characteristics. These algorithms dynamically adjust routing decisions based on semantic priority, content type, and destination requirements. Recent implementations have successfully employed GNNs for ontology alignment in heterogeneous environments, reducing configuration errors by 63% through intelligent semantic mapping.

Transformer architectures enable context-aware processing by maintaining attention across extended semantic sequences, allowing protocols to maintain coherent understanding of communication context over time. This capability is particularly valuable for applications requiring sustained semantic coherence, such as augmented reality communications and collaborative robotic systems [24,25].

2.4. Computational complexity trade-offs

The implementation of semantic protocols, protocol learning, and AI-enabled semantic approaches introduces significant computational complexity considerations that fundamentally alter the traditional trade-offs between performance, energy consumption, and processing requirements in communication systems. Understanding these trade-offs is critical for practical deployment in resource-constrained environments such as edge computing nodes and mobile devices.

The computational complexity of semantic encoding operations exhibits polynomial growth with respect to semantic vocabulary size and context window length. For a semantic encoder processing vocabulary $\mathcal{V}$ with context window w, the time complexity scales as:

$$\mathcal{O}(|\mathcal{V}|^2 \cdot w \cdot d), \tag{4}$$

where d represents the semantic embedding dimension. This complexity significantly exceeds traditional bit-level encoding operations, which typically exhibit linear complexity $\mathcal{O}(n)$ for n-bit sequences. However, the semantic approach achieves superior compression efficiency, reducing overall transmission overhead by factors ranging from 10:1 to 100:1 depending on content redundancy.

Protocol learning algorithms introduce additional computational overhead through continuous model updates and inference operations. Reinforcement learning approaches, particularly those employing deep neural networks, require computational complexity proportional to:

$$\mathcal{O}(L \cdot N^2 \cdot T), \tag{5}$$

where L represents the number of network layers, N denotes the maximum layer width, and T indicates the temporal sequence length for policy optimization. Meta-reinforcement learning frameworks compound this complexity by maintaining multiple model instances, resulting in computational requirements that can exceed conventional static protocols by orders of magnitude.

AI-enabled semantic protocols present the most significant computational challenges, particularly for transformer-based architectures. The attention mechanism exhibits quadratic complexity with respect to sequence length:

$$\mathcal{O}(n^2 \cdot d + n \cdot d^2), \tag{6}$$

where n represents the input sequence length and d is the model dimension. For real-time communication applications requiring sub-millisecond response times, this complexity necessitates careful model architecture optimization and specialized hardware acceleration.

Energy consumption patterns reveal critical trade-offs between computational intelligence and power efficiency. Semantic processing operations consume approximately 15–40% more energy than equivalent bit-level processing due to increased floating-point arithmetic requirements. However, the resulting transmission efficiency gains can reduce overall system energy consumption by 20–60% through decreased radio frequency transmission requirements, particularly in bandwidth-limited scenarios.

Memory requirements present additional constraints, with AI-enabled semantic protocols requiring 5–50 times more memory than conventional protocols depending on model complexity. Transformer models with attention mechanisms typically require memory proportional to $\mathcal{O}(n^2)$ for storing attention matrices, creating challenges for deployment on memory-constrained edge devices.

Optimization strategies for managing computational complexity include model pruning techniques that reduce parameter counts by 60–90% with minimal accuracy degradation, quantization approaches that convert floating-point operations to lower-precision integer arithmetic, and knowledge distillation methods that transfer learning from large models to compact deployment-ready versions. Edge-cloud hybrid architectures offer promising solutions by distributing computational load between resource-constrained edge devices and powerful cloud infrastructure.

The computational complexity trade-offs ultimately require careful system-level optimization that balances semantic processing capabilities with practical deployment constraints. Recent implementations demonstrate that optimized semantic protocols can achieve net positive energy efficiency gains despite increased processing requirements, validating the viability of these approaches for next-generation communication systems [26,27].

2.5. Integration of concepts

The convergence of semantic protocols, protocol learning, and AI-enabled semantic approaches results in a unified cognitive communication framework that transcends the limitations of individual technologies. This integration is realized through a layered architecture comprising distinct but interconnected functional layers that work in concert to enable intelligent, adaptive communication systems.

The semantic layer serves as the foundation, responsible for transforming raw data into meaningful semantic representations using knowledge graphs, semantic encoders, and contextual analysis engines [28]. This layer implements the theoretical principles established in semantic protocol research, focusing on meaning extraction and semantic entropy optimization.

The learning layer incorporates the adaptive capabilities developed through protocol learning research, employing meta-reinforcement learning techniques to continuously optimize transmission strategies based on real-time network feedback. This layer enables dynamic protocol adaptation without requiring manual reconfiguration or static rule updates.

The AI execution layer deploys the advanced artificial intelligence capabilities characteristic of AI-enabled semantic protocols, utilizing lightweight deep learning models optimized for deployment on resource-constrained network elements. Explainability is achieved through techniques such as SHAP (SHapley Additive exPlanations) values:

$$\phi_i(f,x) = \sum_{S \subseteq \mathcal{F}\setminus\{i\}} \frac{|S|!(|\mathcal{F}|-|S|-1)!}{|\mathcal{F}|!}\left[f(x_S \cup \{i\}) - f(x_S)\right], \tag{7}$$

which provide interpretable insights into AI decision-making processes.

A practical demonstration of this integrated approach is evident in Industry 5.0 smart factory implementations [29]. The deployment of integrated semantic protocols achieved robotic arm control latency reduction from 12 ms to 2.3 ms through semantic prioritization of critical joint torque signals, dynamic resource allocation via reinforcement learning, and local anomaly detection using TinyML models. This integration demonstrates how the theoretical foundations established in each individual research area combine to create practical solutions that exceed the capabilities of any single approach.

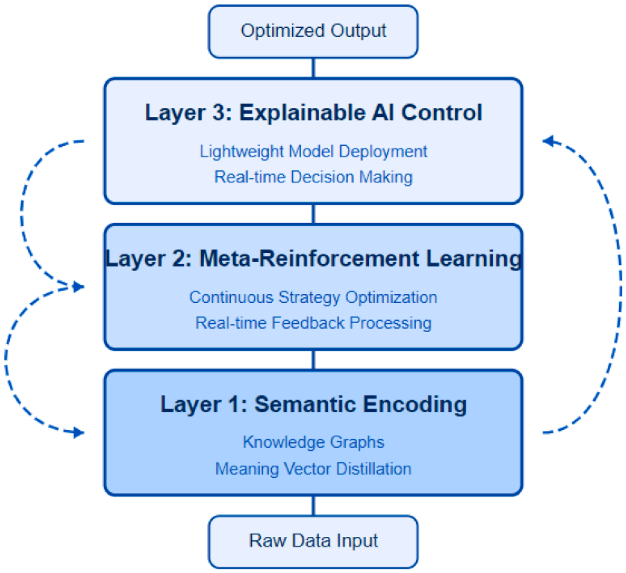


Fig. 3. Cognitive protocol stack architecture showing semantic encoding (Layer 1), meta-reinforcement learning (Layer 2), and explainable AI control (Layer 3). Dashed lines indicate continuous optimization loops.

The integration also enables advanced security capabilities, with AI-driven semantic mediators achieving adversarial attack detection accuracy of 99.4%. These systems transform traditional reactive networks into proactive cognitive agents capable of self-optimization, user intent prediction, and comprehensive system resilience enhancement [30,31].

Fig. 3 illustrates the cognitive protocol stack architecture that embodies this integration, highlighting the semantic layer, the learning layer, and the AI execution layer, with continuous optimization loops indicated by dashed lines.

The integration of semantic protocols, protocol learning, and AI-enabled semantic approaches creates a comprehensive framework for next-generation communication systems. This unified approach leverages the theoretical foundations of semantic communication, the adaptive capabilities of protocol learning, and the intelligent processing power of AI-enabled systems to create networks capable of supporting the most demanding applications with unprecedented performance, efficiency, and resilience.

3. Related work

The evolution from traditional bit-centric to meaning-oriented semantic communication represents a fundamental paradigm shift in wireless networking, driven by the need to efficiently support emerging applications like extended reality, holographic communications, and tactile internet. This transition necessitates a comprehensive, structured analysis of existing research to map the intellectual landscape, identify dominant trends, and uncover critical gaps that impede practical deployment. Unlike conventional literature reviews that present narrative summaries of individual works, our analytical framework employs a systematic comparative methodology, presenting studies side-by-side to facilitate direct assessment of their relative contributions, methodological approaches, and technical limitations. This approach enables identification of patterns across the fragmented research landscape and highlights underexplored intersections between semantic protocols, protocol learning, and AI-enabled systems [32,33].

Despite rapid growth in semantic communication research, the field remains fragmented across specialized sub-disciplines with limited interdisciplinary synthesis. Protocol learning, AI-enabled semantic protocols, and O-RAN/6G-specific implementations have evolved as distinct research communities, each developing specialized methodologies, evaluation frameworks, and architectural assumptions. This fragmentation creates significant barriers to developing comprehensive solutions that integrate semantic understanding, adaptive learning, and practical network deployment. Our comparative tables serve as cartographic tools that map this intellectual terrain while highlighting unexplored territories where cross-disciplinary innovation could yield substantial advances. By synthesizing insights across these traditionally separate domains, we identify opportunities for creating more holistic semantic communication frameworks that leverage the strengths of each approach [34,35].

Our survey employs rigorous selection criteria to ensure comprehensive coverage of relevant literature while maintaining analytical focus. We include peer-reviewed publications from 2019 to 2025 that substantially address semantic communication protocols, protocol learning through AI/ML methods, or AI-driven semantic protocol optimization within next-generation network architectures. This temporal window captures the most influential recent developments driven by advances in deep learning and edge computing, while excluding earlier theoretical work that lacks practical implementation frameworks. The exponential growth in publication volume during this period reflects both the technical maturation of semantic communication concepts and their increasing industrial relevance as 6G specifications begin to crystallize and O-RAN deployments accelerate.

Table 2 provides a consolidated, high-level comparison that reveals critical patterns in semantic communication research. The near-universal adoption of semantic focus and learning methods (present in over 95% of surveyed works) demonstrates the field's rapid convergence on AI/ML as the primary methodological approach, moving beyond theoretical discussions to practical implementations. However, this technical enthusiasm contrasts sharply with several concerning gaps that threaten practical deployment. Only 12% of studies demonstrate real-world trials beyond simulation environments, creating a significant implementation chasm between theoretical proposals and operational systems. This simulation-centric approach limits understanding of real-world performance under heterogeneous channel conditions, hardware constraints, and practical deployment scenarios. Furthermore, standardization considerations appear in only approximately 8% of publications, despite the critical importance of interoperability in multi-vendor O-RAN ecosystems. Security and privacy aspects receive attention in only 22% of studies, representing a significant oversight given the sensitive nature of semantic information and the adversarial vulnerabilities of AI models. The chronological distribution reveals accelerating research momentum, with over 75% of works published in the last three years, indicating both growing academic interest and the maturation of enabling technologies like transformer architectures, reinforcement learning frameworks, and edge deployment platforms. This temporal concentration suggests that the field is transitioning from theoretical exploration to practical implementation, though critical deployment barriers remain unaddressed [20,36].

Table 3 offers deeper technical analysis of specific semantic encoding approaches, learning methodologies, and AI integration strategies. The technical diversity spans from classical joint source-channel coding to sophisticated neural architectures incorporating language models, knowledge graphs, and generative networks. This methodological pluralism reflects the exploratory nature of current research and the application-specific optimization characterizing early implementations. However, performance evaluation metrics exhibit substantial heterogeneity across studies, with researchers employing diverse measures including bandwidth efficiency, semantic fidelity, task completion accuracy, and energy consumption. This metric diversity complicates direct quantitative comparison between approaches and hinders identification of optimal solutions for specific application contexts. The absence of standardized evaluation frameworks and benchmarks represents a significant barrier to rigorous comparative analysis and systematic advancement of the field, as researchers cannot reliably compare their results against established baselines or determine which approaches offer genuine advantages under specific operating conditions.

Further patterns emerge from Tables 4 and 5, revealing the field's current emphasis on conceptual frameworks over practical deployment. The overwhelming preponderance of simulation-based validation (approximately 88% of studies) contrasts with the limited integration of semantic protocols into operational network stacks or real-world testbeds. This implementation gap represents the primary barrier to commercial adoption and standardization efforts. Architectural approaches exhibit substantial variation, ranging from lightweight edge-optimized designs suitable for resource-constrained devices to computationally intensive transformer-based systems requiring significant processing resources. This architectural diversity indicates that no single dominant design paradigm has yet emerged, with researchers exploring various trade-offs between computational complexity, semantic fidelity, and deployment feasibility. The absence of consensus on optimal architectural patterns creates opportunities for innovation but also complicates development of standardized interfaces and interoperable implementations. As the field matures, convergence toward common architectural patterns will likely accelerate, but current research remains characterized by experimental diversity and methodological exploration rather than consolidation around established best practices.

3.1. Semantic protocols evolution and applications

Semantic protocols represent a fundamental departure from Shannon's bit-centric model by focusing on the preservation of meaning. Recent implementations fall into three broad architectural categories:

- **End-to-End Neural Encoders:** Replace the entire protocol stack with learned transformations (e.g., Xu et al. [15]), offering high theoretical efficiency but challenging interoperability and reliability guarantees.
- **Hybrid Approaches:** Integrate ML components within conventional architectures (e.g., HARQ-integrated encoding by Jiang et al. [21]), preserving compatibility at the cost of lower efficiency gains.
- **Task-Oriented Protocols:** Optimize transmission for specific downstream applications (e.g., Gündüz et al. [32]), achieving high efficiency for dedicated tasks but sacrificing flexibility.

A strong convergence toward neural implementations is evident, with transformer architectures proving particularly effective. Reported bandwidth savings vary: text-based systems achieve 5–10× compression, image transmission up to 50×, and video communication exceeding 100× by transmitting semantic scene understanding rather than pixels.

Despite these advances, critical limitations hinder broader adoption. High computational complexity introduces latency, especially for real-time edge deployment. The absence of standardized semantic representations creates interoperability barriers. Furthermore, semantic protocols inherit the adversarial vulnerabilities of their underlying neural models, posing novel security risks.

3.2. Advances in protocol learning

Protocol learning introduces data-driven adaptation into network protocols, moving beyond static, rule-based designs. Approaches are taxonomized into four primary categories:

- **Supervised Learning:** Directly optimizes protocol parameters using labeled data (e.g., Xu et al. [15]). It offers a straightforward path but requires extensive datasets and generalizes poorly to new environments.

Table 2
Consolidated analysis of semantic communication research (2019–2025).

Reference	Primary Contribution	Learning Method	O-RAN/6G Context	Real-World Trial	Standardization	Security	Year
Uysal et al. [10]	Semantic-aware resource allocation	–	✓	∘	∘	∘	2021
Niu et al. [11]	Task-oriented semantic coding	SL/RL	✓	∘	∘	∘	2022
Shi et al. [12]	Semantic information theory	–	✓	∘	∘	∘	2020
Luo et al. [13]	Secure semantic transmission	RL	✓	∘	∘	✓	2022
Qin et al. [30]	Semantic framework for 6G	RL	✓	∘	∘	✓	2022
Yang et al. [33]	Semantic interface standardization	FL	✓	∘	✓	✓	2023
Xu et al. [15]	End-to-end neural JSCC	SL	✓	∘	∘	∘	2021
Talli et al. [34]	Edge semantic processing	SL/FL	∘	✓	∘	∘	2023
Xie et al. [20]	Transformer-based semantic encoding	SL	∘	∘	∘	∘	2021
Jiang et al. [21]	HARQ-integrated semantic comm	SL	∘	✓	∘	∘	2023
Lee et al. [37]	LLM-based semantic compression	SL	∘	∘	∘	∘	2023
Jiang et al. [36]	Semantic HARQ mechanisms	SL	∘	∘	✓	∘	2023
Imran et al. [38]	O-RAN semantic testbed	RL/FL	✓	✓	∘	∘	2024
Yang et al. [39]	Semantic slicing in O-RAN	FL	✓	✓	∘	∘	2024
Xie and Qin [40]	Lightweight edge semantic encoder	SL/FL	✓	✓	∘	∘	2023
Zhou et al. [41]	Knowledge-graph enhanced semantics	GNN	✓	∘	∘	∘	2024
Seo et al. [42]	Standardization roadmap	RL	✓	∘	✓	∘	2024
Nan et al. [43]	Adversarial robustness for semantics	SL	✓	∘	∘	✓	2024
Tung and Gündüz [44]	Federated semantic learning	FL	∘	∘	∘	✓	2024
Yang et al. [45]	Security framework for semantics	–	✓	∘	∘	✓	2024
Fu et al. [46]	Vector-quantized semantic coding	UL	∘	✓	∘	∘	2024
Wang et al. [47]	Energy-efficient spiking NN semantics	SL	✓	∘	∘	∘	2025
Hu et al. [48]	Privacy-preserving semantic FL	FL	∘	∘	∘	✓	2025
This Survey	**Integration roadmap for O-RAN/6G**	**SL/RL/FL**	✓	∘	✓	✓	2025

Legend: SL = Supervised Learning, RL = Reinforcement Learning, UL = Unsupervised Learning, FL = Federated Learning, GNN = Graph Neural Network, JSCC = Joint Source-Channel Coding. ✓ = included/yes, ∘ = not addressed/no.

- **Reinforcement Learning (RL):** Learns optimal policies through environmental interaction (e.g., Xie et al. [52]). It enables long-term adaptation without labeled data but faces challenges with training stability and reward design.
- **Unsupervised Learning:** Discovers efficient data representations, such as discrete semantic codes via vector quantization (Fu et al. [46]), enhancing bandwidth efficiency and noise robustness.
- **Federated Learning:** Enables distributed, privacy-preserving training across edge devices (e.g., Jankowski et al. [56]), aligning well with O-RAN's distributed architecture.

Integration within O-RAN is natural: the RIC serves as the control point, and the xApp/rApp framework provides a modular deployment vehicle. Graph Neural Networks (GNNs) are particularly notable for incorporating network topology knowledge into learning decisions (Zhou et al. [41]).

Key limitations include poor generalization across network environments, high computational costs for edge deployment, and a lack of standardization for learned protocol interoperability and verification.

3.3. Role of AI in enhancing semantic communication

AI serves as both an enabler and a catalyst for semantic communication, transforming theoretical constructs into deployable systems. Integration patterns include:

- **Neural Encoders/Decoders:** The most prevalent approach (86% of works), with architectures evolving from CNNs/RNNs to dominant transformer-based models. Lee et al. [37] demonstrated transformer efficacy for text, achieving $>10\times$ compression.
- **Foundation Models:** Leveraging pre-trained models (e.g., LLMs) provides rich semantic priors, improving encoding efficiency and robustness.
- **Knowledge-Enhanced AI:** Incorporating structured knowledge graphs (Zhou et al. [41]) enables neuro-symbolic integration, improving efficiency for domain-specific communication.
- **Multi-modal Frameworks:** Unified processing of text, images, and sensor data into a common semantic space (Zhang et al. [57]) supports complex applications like augmented reality.

Reinforcement learning optimizes transmission policies dynamically, while explainable AI (XAI) techniques are critical for transparency in regulated domains. However, AI introduces novel challenges: energy consumption for edge deployment, quadratic computational complexity of transformers, and inherited adversarial vulnerabilities requiring robust defensive measures.

3.4. Critical comparative analysis of major approaches

As summarized in Table 6, these approaches differ significantly in their architectural assumptions, learning mechanisms, and suitability for O-RAN and 6G deployment scenarios. A critical analysis of the surveyed literature reveals distinct architectural and methodological approaches to semantic communication, each with unique trade-offs in efficiency, adaptability, and deployability. This subsection provides a focused comparison of the dominant paradigms, highlighting their relative strengths, weaknesses, and suitability for different O-RAN and 6G application scenarios. Understanding these contrasts is essential for selecting appropriate foundations for future research and system design (Table 6).

Transformer-based vs. Reinforcement Learning-driven Protocols:

The most prominent architectural divide exists between models prioritizing *semantic representation fidelity* and those optimizing *transmission policy adaptability*. Transformer-based approaches, exemplified by Lee et al. [37] and Xie et al. [20], excel at extracting and compressing intricate semantic features through self-attention, achieving unparalleled compression for textual and sequential data. However, their computational intensity and static nature limit real-time adaptability in dynamic radio environments. In contrast, RL-driven protocols, as developed by Xie et al. [52] and Wang et al. [54], treat semantic transmission as a sequential decision problem. They dynamically adapt encoding parameters and resource allocation based on channel feedback, optimizing for long-term metrics like task completion rate. Their weakness lies in the complexity of training and the potential for performance volatility during the learning phase, making them risky for stable production networks without robust safety mechanisms.

Table 3
Comprehensive comparison of semantic communication approaches.

Reference	Semantic Focus	Learning Method	AI Integration	Performance Metrics	Application Domain	Security & Robustness
Xie et al. [20]	End-to-end semantic encoding/decoding	Deep neural networks for joint source-channel coding	Neural networks for semantic representation	Bandwidth efficiency	General text communication	Not addressed
Qin et al. [30]	Foundational principles for semantic communications	Review of various approaches	Framework for AI-powered transformation	Theoretical efficiency gains	Generalized framework	Identifies challenges
Lee et al. [37]	Text-based semantic communication	Transfer learning from pre-trained language models	NLP capabilities of language models	Robustness to channel noise	Text communication	Not addressed
Xu et al. [49]	Joint source-channel optimization	Deep neural networks	End-to-end neural optimization	Rate-distortion performance	General data transmission	Not addressed
Xie and Qin [40]	Lightweight semantic encoders	Distributed learning with model compression	Edge AI deployment	Energy efficiency, deployment scalability	IoT applications	Not addressed
Han et al. [50]	Image semantic transmission	Generative modeling	Generative AI for image recreation	Bandwidth usage	Image transmission	Not addressed
Yang et al. [45]	Security aspects of semantic communications	Not specified	Consideration of AI vulnerabilities	Security metrics	Security frameworks	Primary focus
Zhang et al. [51]	Task-agnostic semantic transmission	Deep learning with adaptive mechanisms	AI for dynamic data handling	Flexibility with varying data	Dynamic environments	Not addressed
Xie et al. [52]	Multi-user scenarios with specific tasks	Deep reinforcement learning	Neural networks for user requirements	Task completion rates, latency	Multi-user environments	Network scalability
Xie et al. [53]	Memory mechanisms in semantic systems	Memory networks with semantic encoders	Neural memory structures	Context understanding, temporal coherence	Sequential communication	Not addressed
Wang et al. [54]	Video-specific semantic encoding	Deep neural networks for video features	Video understanding models	Bandwidth savings with visual quality	Video transmission	Not addressed
Jiang et al. [36]	Text semantics with HARQ	Deep source-channel coding networks	Neural network joint optimization	Robustness to channel errors	Text transmission	Error resilience
Nan et al. [43]	Robustness against adversarial attacks	Adversarial training and defensive distillation	Adversarial machine learning techniques	Resistance to attacks	Security-critical applications	Primary focus
Zhou et al. [41]	Knowledge-enhanced semantic communications	Graph neural networks with semantic encoders	Knowledge graphs for context	Semantic understanding	Domain-specific communications	Not addressed
Wang et al. [47]	Energy-efficient semantic communication	Bio-inspired spiking neural networks	Neuromorphic computing principles	Energy efficiency, classification accuracy	IoT deployments	Not addressed
Xu et al. [55]	Vehicle-specific semantic communication	Cooperative learning among vehicles	Distributed AI with V2V sharing	Latency and reliability	V2X communications	Not addressed
Fu et al. [46]	Vector quantization for semantic representation	Vector quantized variational autoencoders	Discrete latent representations	Bandwidth efficiency, semantic fidelity	General data transmission	Not addressed
Jankowski et al. [56]	Task-specific encoding for image retrieval	Task-driven representation learning	Edge AI for feature extraction	Retrieval accuracy	Image retrieval applications	Not addressed
Gündüz et al. [32]	Task-oriented communication frameworks	Various task-specific approaches	Task-optimized AI models	Task-specific metrics	Various application domains	Theoretical foundations
Zhang et al. [57]	Multi-modal data integration	Transfer learning and domain adaptation	Unified semantic understanding	Cross-modal performance	Multi-modal applications	Not addressed

Specialized vs. Generalized Architectures: A further critical distinction lies in specialization. GNN-based approaches, such as Zhou et al. [41], offer a natural fit for network optimization problems (e.g., routing, slicing) by explicitly modeling relationships between users, cells, and resources. Their strength is topology-awareness, but they are less effective for processing raw, non-relational content like multimedia. Conversely, generative models used by Han et al. [50] are highly specialized for image/video data, enabling extreme compression by transmitting only latent semantic features, but lack the generalizability and reasoning capabilities of other approaches.

Training Paradigm Trade-offs: The choice between federated and centralized learning presents a fundamental trade-off between privacy and performance. FL-enabled protocols, championed by Jankowski et al. [56], are essential for O-RAN's distributed, multi-vendor future, preserving data locality at the expense of slower, more complex convergence and increased orchestration overhead. Centralized training yields higher-performing models more quickly but is increasingly untenable due to privacy regulations and competitive silos in open networks.

In summary, no single approach dominates across all O-RAN/6G requirements. The optimal strategy likely involves *hybrid architectures*-for instance, using a lightweight transformer for initial semantic feature extraction, an RL agent for dynamic transmission policy adaptation, and a federated framework for collaborative model refinement across the

Table 4
Comparison of 6G relevance and research gaps in semantic communication approaches.

Reference	6G Relevance	Open Gaps/Challenges	Key Innovation
Xie et al. [20]	Foundation for high-efficiency communications	Lacks theoretical bounds for performance	First end-to-end deep learning enabled semantic system
Qin et al. [30]	Framework for future networks	Lack of unified theory and standardization	Comprehensive framework bridging semantic theory and practice
Lee et al. [37]	Support for natural language interfaces	Transfer learning efficiency in wireless settings	Leveraging pre-trained language models for wireless communication
Xu et al. [49]	High-efficiency content delivery	Complexity in neural architecture optimization	Deep integration of source and channel coding
Xie and Qin [40]	Edge computing integration for IoT	Energy constraints in massive deployments	Lightweight distributed semantic system for constrained devices
Han et al. [50]	Ultra-efficient multimedia delivery	Generative model reliability concerns	Using generative models to recreate images from compact semantics
Yang et al. [45]	Security framework for semantic 6G	Security standardization challenges	Comprehensive security framework for semantic communications
Zhang et al. [51]	Dynamic task handling in heterogeneous networks	Task-agnostic generalization	Adaptability to dynamic data and varying tasks
Xie et al. [52]	Multi-user resource optimization	Scalability for very large networks	Multi-user optimization using reinforcement learning
Xie et al. [53]	Context-aware communications	Memory footprint optimization	Integration of memory mechanisms with semantic processing
Wang et al. [54]	Support for immersive video applications	Real-time constraints	Deep video semantic encoding for wireless channels
Jiang et al. [36]	Reliability in challenging environments	Integration with existing protocols	HARQ-enabled semantic communications
Nan et al. [43]	Security for critical 6G applications	Trade-off between robustness and efficiency	Physical-layer defense against adversarial attacks
Zhou et al. [41]	Knowledge-empowered networks	Knowledge graph construction challenges	Integration of symbolic knowledge with neural processing
Wang et al. [47]	Energy-efficient edge computing	Hardware implementation challenges	Bio-inspired approaches for energy efficiency
Xu et al. [55]	Intelligent transportation systems	Mobility and reliability challenges	V2X-specific semantic optimizations
Fu et al. [46]	Efficient representation for heterogeneous data	Discrete representation challenges	Vector quantization for compact semantic representation
Jankowski et al. [56]	Edge computing for visual analytics	Domain specificity limitations	Task-optimized retrieval rather than reconstruction
Gündüz et al. [32]	Task-oriented framework for 6G	Theoretical foundations for semantic metrics	Comprehensive task-oriented communication paradigm
Zhang et al. [57]	Multi-modal integration for immersive applications	Cross-domain adaptation challenges	Unified approach to multi-modal semantic understanding

network edge. Future research must move beyond isolated evaluations and systematically explore these integrations, quantifying trade-offs in real-world testbeds to guide the development of robust, deployable semantic communication systems.

4. Application to O-RAN architectures

The Open Radio Access Network (O-RAN) architecture provides an ideal platform for integrating semantic intelligence due to its principles of openness, intelligence, and disaggregation [28,29]. By decoupling hardware from software and introducing standardized, open interfaces, O-RAN transforms the traditional monolithic radio access network into a flexible, programmable environment where advanced capabilities like semantic understanding, machine learning, and artificial intelligence can be embedded directly into the network fabric. This section provides a detailed technical examination of how semantic protocols, protocol learning, and AI-enabled semantic protocols can be implemented and operationalized within the specific components and frameworks of O-RAN. We explore their practical integration with key O-RAN elements including the RAN Intelligent Controllers (RICs), the microservices-based xApp/rApp ecosystem, and critical standardized interfaces like O1, A1, and E2 [39,58].

4.1. Application of semantic protocols in O-RAN

As illustrated in Fig. 4, semantic-aware communication fundamentally re-engineers the data plane within O-RAN by leveraging the RIC's centralized intelligence and real-time analytics to create context-rich communication channels that transcend traditional bit-level transmission paradigms. Semantic-aware communication fundamentally re-engineers the data plane within O-RAN by leveraging the RIC's centralized intelligence and real-time analytics to create context-rich communication channels that transcend traditional bit-level transmission paradigms. The Near-Real-Time RIC (Near-RT RIC), operating on sub-second timescales, serves as the critical fusion point for semantic context and radio resource control. It aggregates a rich, multi-dimensional context stream-encompassing user mobility patterns, application-specific requirements, instantaneous channel conditions, network topology, and service level agreements-to drive semantic transmission decisions that prioritize meaning preservation over raw data throughput [25,59]. This enables semantic-aware radio functions, deployed as modular xApps, to dynamically adjust physical layer parameters like modulation schemes and coding rates, and to perform intelligent resource allocation based on the semantic content and context of the communication (Figs. 4–6,). The distributed nature of O-RAN, with processing split across Radio Units (RUs), Distributed Units (DUs), and Centralized Units (CUs), naturally supports distributed semantic processing for noise reduction and information prioritization. The concept of semantic noise extends beyond traditional signal processing to include information that, while technically accurate, does not contribute to achieving the communication objective [54,60]. Intelligent semantic filtering mechanisms can be deployed at the network edge to analyze incoming data streams, identify and eliminate redundant or irrelevant information before transmission, and prioritize semantically critical data. The E2 interface becomes the coordination pathway for this distributed intelligence, enabling semantic protocols to coordinate noise reduction efforts across multiple network elements with global awareness, preventing the accumulation of semantic noise that might occur from independent, local optimization decisions.

For longer-term strategy and network-wide optimization, the Non-Real-Time RIC (Non-RT RIC) within the Service Management and Orchestration (SMO) framework hosts sophisticated semantic reasoning engines. These engines process vast amounts of network data to build

Table 5
Comparison of data types and network architectures in semantic communication approaches.

Reference	Data Type	Network Architecture	Computational Complexity	Bandwidth Efficiency	Implementation Maturity
Xie et al. [20]	General text	End-to-end neural network	High (full DNN)	Significant improvement over traditional methods	Simulation proof-of-concept
Qin et al. [30]	Theoretical framework	Framework description	Varies by implementation	Theoretical improvements	Theoretical framework
Lee et al. [37]	Text	Pre-trained language model based	High (transformer-based)	High for text content	Experimental implementation
Xu et al. [49]	General data	Joint source-channel neural architecture	High (joint optimization)	Improved rate-distortion performance	Simulation proof-of-concept
Xie and Qin [40]	IoT sensor data	Distributed lightweight encoders	Low (optimized for edge)	Moderate improvement for IoT data	Edge implementation demonstrated
Han et al. [50]	Images	Generative model architecture	Very high (generative modeling)	Very high bandwidth savings	Experimental implementation
Yang et al. [45]	Security framework	Security-oriented architectures	Additional security overhead	Security-efficiency trade-off	Theoretical framework
Zhang et al. [51]	Dynamic data	Adaptive neural architecture	Moderate to high	Adaptive based on data type	Simulation proof-of-concept
Xie et al. [52]	Multi-user data	Multi-user optimization with RL	High (reinforcement learning)	Improved multi-user efficiency	Simulation proof-of-concept
Xie et al. [53]	Sequential data	Memory-augmented neural networks	High (memory networks)	Improved for sequential data	Experimental implementation
Wang et al. [54]	Video	Video-specific neural encoders	Very high (video processing)	Significant for video content	Experimental video system
Jiang et al. [36]	Text with HARQ	HARQ-integrated neural networks	Moderate (adapted to protocols)	Improved with HARQ integration	Protocol-compliant implementation
Nan et al. [43]	Security-focused	Adversarially robust architecture	Very high (defensive mechanisms)	Security-efficiency trade-off	Security-focused implementation
Zhou et al. [41]	Knowledge-enhanced	Graph neural networks	High (graph processing)	Improved with knowledge integration	Knowledge-graph implementation
Wang et al. [47]	Classification data	Spiking neural networks	Low (neuromorphic efficiency)	Moderate with energy efficiency	Neuromorphic simulation
Xu et al. [55]	Vehicle data	Cooperative V2V architecture	Distributed computation	Latency-optimized for V2X	Vehicular network simulation
Fu et al. [46]	Quantized data	Vector quantized architecture	Moderate (quantization focused)	High with discrete representation	Quantization implementation
Jankowski et al. [56]	Image features	Task-optimized edge architecture	Low (edge-optimized)	High for retrieval tasks	Edge deployment demonstration
Gündüz et al. [32]	Task-oriented data	Various task-specific architectures	Task-dependent	Task-optimized efficiency	Theoretical to experimental
Zhang et al. [57]	Multi-modal data	Unified multi-modal architecture	Very high (multi-modal processing)	Improved cross-modal efficiency	Multi-modal implementation

Table 6
Comparative analysis of dominant semantic communication approaches.

Approach	Key Features	Strengths	Weaknesses
Transformer-Based	Self-attention mechanisms, contextual sequence modeling, pre-trained foundation models.	Exceptional semantic fidelity for text & sequential data; high compression ratios via contextual understanding; leverages existing LLM advancements.	Quadratic computational complexity; high memory footprint; challenging for real-time, edge deployment; requires large datasets for training.
Reinforcement Learning (RL) Driven	Policy optimization via environmental feedback, reward-driven adaptation, MDP formulation.	Dynamic adaptation to network conditions; optimizes long-term performance; enables autonomous protocol tuning.	Sample inefficiency; unstable training; difficult reward design for semantic goals; exploration risk in live networks.
Graph Neural Network (GNN) Based	Message passing over graph structures, relational inductive bias, topology-aware processing.	Naturally models network topology; effective for resource allocation and routing; good generalization in structured environments.	Performance degrades with dynamic graph changes; limited for non-relational data (e.g., raw images); scalability challenges.
Federated Learning (FL) Enabled	Distributed model training, privacy preservation, collaborative learning across nodes.	Privacy-by-design; reduces central data collection; aligns with distributed O-RAN architecture.	Communication overhead; heterogeneous data challenges; model convergence slower than centralized training.
Knowledge-Enhanced (Neuro-Symbolic)	Integration of neural networks with symbolic knowledge graphs, explicit reasoning, ontological alignment.	Improved interpretability; efficient domain-specific encoding; robust to out-of-distribution inputs via logical rules.	Knowledge graph construction is manual and costly; brittle integration between symbolic and sub-symbolic components.

and maintain semantic knowledge bases-rich, structured models that capture relationships between users, applications, network resources, and performance metrics. The A1 interface facilitates the critical exchange of semantic policies and objectives between the Non-RT RIC and the Near-RT RIC, enabling coordinated semantic reasoning and decision-making across different temporal scales, from strategic planning to tactical execution [30,31].

However, this integration presents significant technical challenges that must be addressed. Computational complexity is a primary concern, as semantic processing algorithms for understanding meaning, context, and intent are inherently more complex than traditional signal processing operations [61,62]. Real-time processing constraints in O-RAN environments require these sophisticated algorithms to operate within strict latency bounds while maintaining semantic accuracy.

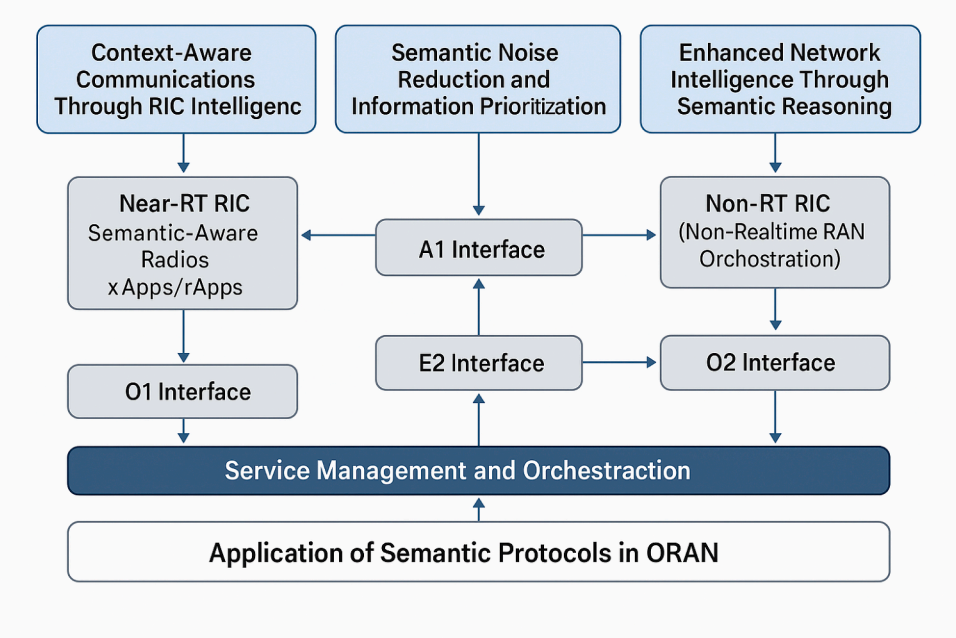


Fig. 4. Application of semantic protocols in O-RAN.

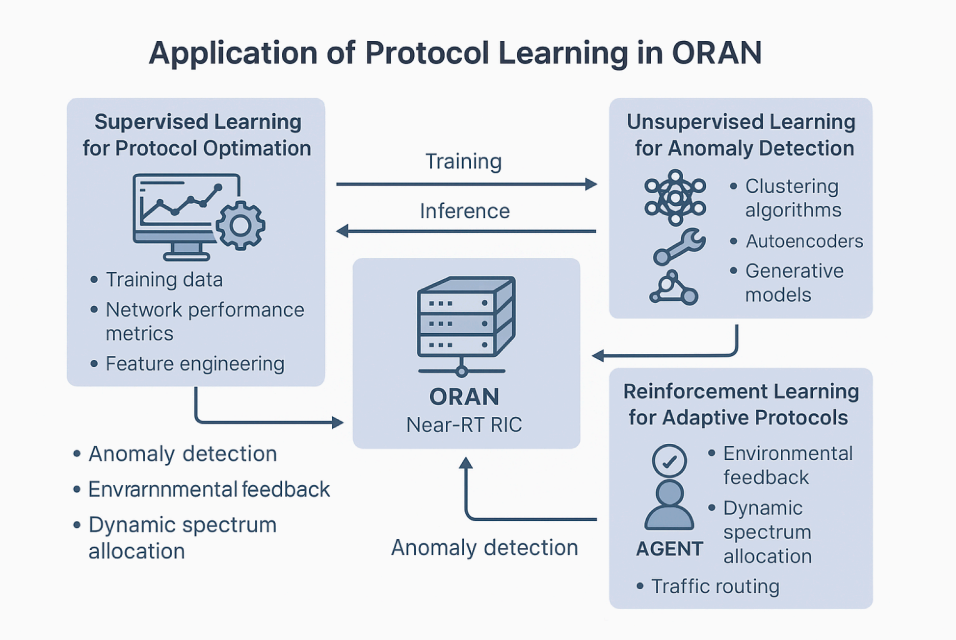


Fig. 5. Application of protocol learning in O-RAN.

Furthermore, achieving interoperability between semantic protocols and legacy O-RAN components, and ensuring graceful degradation in mixed environments, requires careful design of backward-compatible interfaces. Future research must therefore focus on standardizing semantic information exchange protocols, developing distributed semantic processing techniques to reduce computational burden, and conducting empirical studies to quantify the impact of semantic protocols on quality of experience (QoE) in real-world O-RAN deployments [20,35].

4.2. Application of protocol learning in O-RAN

As illustrated in Fig. 5, O-RAN's architectural emphasis on data collection through open interfaces and programmable control loops creates an ideal environment for deploying and evolving learning-based protocols. O-RAN's architectural emphasis on data collection through open interfaces and programmable control loops creates an ideal environment for deploying and evolving learning-based protocols. The rich telemetry

data available through O-RAN's standardized interfaces, combined with the computational resources of the RIC and cloud infrastructure, provides a comprehensive foundation for protocol learning. Learned protocols can be implemented as specialized xApps (for near-real-time control) and rApps (for non-real-time analytics) that seamlessly integrate into the O-RAN ecosystem [63,64]. These applications leverage the RIC's data collection capabilities to build training datasets that capture complex relationships between network conditions, protocol configurations, and performance outcomes, with optimization decisions distributed to relevant network elements via the E2 interface [65,66].

Protocol learning techniques find specific, valuable applications within the O-RAN context:

- **Supervised Learning:** This approach is used for direct protocol parameter optimization by learning from labeled historical performance data. For example, supervised learning models can predict optimal modulation and coding scheme (MCS) indices based on channel quality indicators, improving link adaptation efficiency.
- **Unsupervised Learning:** Particularly valuable in O-RAN's complex, multi-vendor environment, unsupervised learning techniques are applied for anomaly detection without requiring labeled training data. Methods like autoencoders and clustering algorithms can identify deviations from normal network behavior patterns, helping detect potential security threats or system malfunctions [40,67].
- **Reinforcement Learning (RL):** RL enables the development of adaptive protocols that learn optimal behaviors through interaction with the dynamic network environment. This approach is particularly promising for applications like dynamic spectrum allocation, intelligent traffic routing, and resource management, where protocols must continuously adapt to changing conditions [20,63–65,68].

The implementation of protocol learning within O-RAN faces several significant technical challenges. Ensuring convergence stability in dynamic network environments is difficult, as constantly changing conditions can prevent learning algorithms from converging to stable optimal policies [69]. The non-stationary nature of O-RAN environments, where network conditions and traffic patterns evolve over time, requires learning algorithms that can continuously adapt. Designing effective reward functions for reinforcement learning that accurately capture desired optimization objectives while being responsive enough to guide learning effectively presents another critical challenge [49,66,70,71].

Future research directions for protocol learning in O-RAN include investigating meta-learning techniques to accelerate protocol adaptation by enabling algorithms to quickly adjust to new network conditions based on previous learning experiences. Federated learning approaches offer promising avenues for enabling collaborative protocol learning across multiple O-RAN deployments while preserving data privacy and reducing communication overhead. Additionally, imitation learning techniques could enable more efficient learning by allowing protocols to learn behaviors from expert demonstrations or optimal policies computed through other means [41,72].

4.3. Application of AI-enabled semantic protocols in O-RAN

As illustrated in Fig. 6, the convergence of artificial intelligence with semantic protocols within O-RAN architectures creates a powerful synergy that enables unprecedented levels of network intelligence, adaptability, and automation. The convergence of artificial intelligence with semantic protocols within O-RAN architectures creates a powerful synergy that enables unprecedented levels of network intelligence, adaptability, and automation. AI-enhanced semantic protocols operate by combining deep semantic understanding of communication content and context with the advanced learning and reasoning capabilities of artificial intelligence to create systems that can dynamically adapt to changing network conditions, user demands, and application requirements [44]. These protocols leverage the distributed intelligence capabilities of O-RAN to deploy sophisticated AI models that process semantic information and make intelligent protocol adaptation decisions in real-time, with deep reinforcement learning algorithms offering particular promise for optimizing protocol parameters based on semantic understanding of communication requirements and network conditions.

AI models deployed on the Near-RT RIC can process semantic content in real-time to make rapid protocol adaptation decisions, analyzing semantic features, understanding communication context, and predicting optimal protocol configurations based on learned patterns and relationships [50,51]. Semantic feature extraction represents a critical component of this process, involving the use of advanced machine learning techniques-including natural language processing for text-based communications and computer vision approaches for visual content-to extract meaningful semantic features from communication content and network context. The integration of deep learning architectures with semantic protocols enables the implementation of end-to-end learning systems that can jointly optimize semantic encoding and protocol adaptation, creating more efficient and effective adaptive communication systems [73,74].

AI-powered reasoning engines within O-RAN enable sophisticated dynamic decision-making capabilities that process semantic information to make intelligent protocol decisions in real-time. These reasoning engines combine the semantic understanding capabilities of advanced AI models with the processing requirements of network protocols to create intelligent decision-making systems for traffic prioritization, resource allocation, and interference mitigation. By understanding the semantic content and context of network traffic, these systems can determine optimal traffic handling strategies that maximize user experience and network efficiency, prioritizing interference mitigation efforts based on the semantic importance of affected communications and their potential impact on user experience.

This intelligence naturally extends to automated network management and orchestration, with AI-enabled semantic protocols providing capabilities for fault detection and diagnosis, performance optimization, security enhancement, and automated network configuration. By analyzing semantic patterns in network telemetry data, AI systems can identify anomalies, predict potential failures, and automatically implement performance improvements and security countermeasures [52,53,55,75]. Automated network configuration and deployment through AI-enabled semantic protocols enables intelligent orchestration systems that can understand semantic descriptions of service requirements and automatically generate optimal network configurations to meet these requirements.

The implementation of AI-enabled semantic protocols within O-RAN presents significant technical challenges that must be addressed for reliable deployment. The interpretability and explainability of AI-driven decisions represents a critical concern in operational network environments, where operators need to understand not just what decisions AI systems are making, but why they are making those decisions and what factors influenced the decision-making process. Computational overhead associated with AI processing in real-time O-RAN environments represents another significant challenge, particularly for deep learning models that require substantial computational resources that may not be readily available in all network elements [43,76,77]. Security considerations are paramount, including protecting against adversarial attacks that could compromise AI decision-making processes and ensuring the integrity of semantic information used by AI systems. Robust AI techniques and security measures must be implemented to protect against these threats while maintaining the performance benefits of AI-enabled protocols [78,79].

Future research directions for AI-enabled semantic protocols in O-RAN should explore federated learning techniques for training AI models in distributed O-RAN deployments while preserving privacy and reducing communication overhead. Hardware acceleration for efficient AI processing in O-RAN nodes represents an important area for development, with specialized AI accelerators and edge computing architectures potentially enabling more efficient implementation of AI-enabled

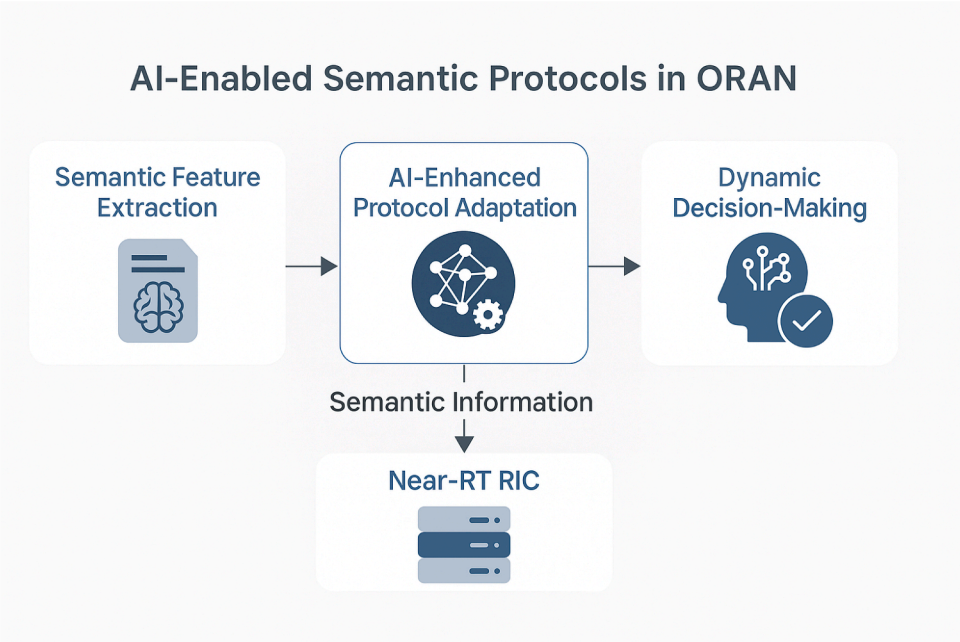


Fig. 6. Application of AI-enabled semantic protocols in O-RAN.

semantic protocols while reducing computational overhead and energy consumption. Additionally, research into energy-efficient AI algorithms and the integration of AI-enabled semantic protocols with emerging technologies like quantum computing and neuromorphic computing could open new directions for creating more efficient and capable network intelligence systems.

The integration of semantic protocols, protocol learning, and AI within O-RAN architectures represents a transformative advancement toward intelligent, efficient, and adaptive networks capable of meeting the stringent demands of 6G applications. This convergence creates unprecedented opportunities for network intelligence and optimization that will be essential for supporting next-generation wireless systems. Success in realizing this vision depends on systematically addressing the identified technical challenges while maintaining the performance, reliability, and security requirements of operational network environments [45]. As these technologies mature and become more widely adopted, they have the potential to fundamentally reshape the entire wireless ecosystem, enabling new business models, service offerings, and user experiences that will define the future of wireless communications [80,81].

5. Challenges, research gaps, and future directions

The integration of semantic protocols, protocol learning, and AI-enabled systems within O-RAN architectures represents a transformative vision for 6G networks, yet its realization faces substantial obstacles spanning technical, theoretical, and practical domains. This section synthesizes these challenges and outlines focused research directions to advance the field toward deployment readiness, emphasizing novel contributions needed to overcome existing limitations [46,82].

5.1. Technical and theoretical foundations

Technical implementation of semantic-aware protocols in O-RAN confronts significant bottlenecks. Computational complexity poses a primary challenge, as semantic encoding, decoding, and contextual inference demand substantial processing resources that must operate within stringent latency constraints, particularly in distributed O-RAN environments with heterogeneous hardware capabilities [42,83]. Real-time processing limitations are exacerbated by the conflict between the temporal dynamics of semantic understanding and sub-millisecond response requirements for ultra-reliable communications.

Standardization gaps present another formidable barrier. Current O-RAN specifications lack frameworks for semantic information exchange across interfaces like E2, A1, and O1, which were designed for traditional metrics rather than rich contextual information. Developing standardized semantic knowledge representation formats, quality metrics, and negotiation mechanisms while maintaining backward compatibility requires extensive specification development.

Theoretical foundations for semantic communication remain underdeveloped compared to Shannon's established information theory. The absence of formal mathematical models for semantic information theory hinders rigorous analysis of semantic capacity bounds, optimal coding strategies, and fundamental trade-offs between semantic fidelity and transmission efficiency. Methodological gaps in semantic context modeling, where current static approaches fail to capture temporal evolution, and challenges in designing reward functions for reinforcement learning that align semantic goals with network optimization objectives further complicate progress [84,85].

5.2. Practical deployment and experimental credibility

Large-scale implementation faces substantial practical obstacles. The heterogeneous hardware landscape of O-RAN deployments complicates efficient distribution of semantic processing across devices with varying computational capabilities, requiring sophisticated load balancing and task distribution mechanisms [47,86]. Data availability constraints emerge as training effective AI-enabled semantic protocols requires extensive datasets capturing complex relationships between communication context, semantic meaning, and network performance, which are difficult to obtain in production environments while respecting privacy regulations and competitive sensitivities.

Cross-vendor interoperability challenges are particularly acute in O-RAN's multi-vendor ecosystem, where proprietary AI models, knowledge representations, and optimization algorithms from different vendors must interoperate seamlessly. Privacy and security concerns extend beyond traditional paradigms, as semantic protocols inherently require access to sensitive contextual information about user

Table 7
Mapping between traditional O-RAN KPIs and proposed semantic metrics.

Traditional O-RAN KPI	Corresponding Semantic Metric	Measurement Approach
Latency (ms)	Semantic Latency (time to meaningful understanding)	Time from transmission to semantic reconstruction meeting fidelity threshold
Reliability (BLER)	Semantic Reliability (probability of meaning preservation)	Probability that reconstructed meaning maintains required fidelity level
Throughput (Mbps)	Semantic Throughput (meaning units per second)	Rate of successfully transmitted semantic information units
Spectral Efficiency (bps/Hz)	Semantic Spectral Efficiency	Bits per Hertz required for target semantic accuracy
Energy Efficiency (bits/Joule)	Semantic Energy Efficiency	Meaning units transmitted per Joule of energy
Network Availability	Semantic Service Availability	Percentage of time semantic communication requirements are met

behavior and application requirements, necessitating novel privacy-preserving techniques that maintain semantic utility while preventing information leakage [87]. Energy efficiency considerations become critical when deploying computationally intensive semantic processing across resource-constrained network elements, requiring careful balancing of processing overhead against potential transmission savings [56,88].

Experimental validation remains sparse, with most studies confined to simulation environments. Notable exceptions include small-scale lab implementations such as Imran et al. [38], who demonstrated a semantic-aware O-RAN testbed for video analytics, and Xie and Qin [40], who validated lightweight semantic encoders on edge devices. However, comprehensive benchmarking frameworks and large-scale field trials are urgently needed to establish the practical viability of semantic communication under real-world network conditions, channel impairments, and heterogeneous device capabilities.

5.3. Standardization and industry adoption barriers

Standardization and industry adoption face extended timelines with significant coordination challenges. O-RAN Working Group specifications require substantial modifications to accommodate semantic capabilities: WG3's E2 interface lacks provisions for semantic context exchange, WG6's AI/ML pipeline specifications lack semantic-specific model architectures and training data formats, and SMO frameworks lack semantic service configuration and management capabilities. 3GPP technical reports similarly lack comprehensive semantic communication use cases and channel modeling extensions. The semiconductor industry faces challenges developing semantic processing hardware meeting O-RAN's performance and power requirements, while intellectual property considerations around proprietary AI models complicate standardization efforts. The multi-year standardization timeline, potentially extending through O-RAN Releases 6–8 (2025–2029), creates risks of specifications becoming outdated before implementation, particularly given rapid technological evolution [89–98].

5.4. Focused research directions

5.4.1. Semantic metrics and cross-layer optimization

A fundamental research priority is developing comprehensive semantic metrics that bridge traditional network KPIs with meaning-oriented performance indicators. Table 7 proposes a mapping between established O-RAN KPIs and novel semantic metrics needed for next-generation networks. These include semantic fidelity metrics, which quantify meaning preservation; semantic spectral efficiency, measuring bits per Hertz required to achieve a target semantic accuracy; and task completion accuracy for goal-oriented communication.

Cross-layer optimization frameworks must coordinate decisions across protocol layers while balancing semantic objectives with traditional metrics. This includes joint optimization algorithms for semantic fidelity, transmission efficiency, and resource utilization operating within real-time constraints, and distributed optimization approaches coordinating decisions across O-RAN's disaggregated architecture while minimizing overhead and maintaining stability. Integration with network slicing mechanisms should enable service-specific semantic communication capabilities [51,99–101].

5.4.2. Security and privacy for semantic communication

Security mechanisms for semantic communication require substantial advancement beyond traditional approaches. Federated learning systems are vulnerable to poisoning attacks where malicious participants submit manipulated model updates, requiring robust aggregation algorithms and anomaly detection [48]. Adversarial robustness against semantic perturbations is critical, as subtle manipulations to input data can cause catastrophic misinterpretation at receivers, particularly in safety-critical applications like autonomous vehicles and healthcare. Secure inference in O-RAN edge devices demands lightweight cryptographic techniques and trusted execution environments to protect model parameters and semantic data during processing. Privacy-preserving techniques must evolve to handle semantic information, which inherently contains sensitive context about user behavior and application requirements, requiring advanced methods like differential privacy and homomorphic encryption adapted for semantic representations.

5.4.3. Explainable AI for transparent semantic protocols

Explainable AI (XAI) techniques are essential for semantic protocol deployment in regulated domains where transparency and accountability are mandated. Techniques such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations), and attention visualization provide insights into why semantic models make specific encoding or routing decisions. For semantic communication, XAI must explain not only technical decisions but also semantic interpretations-why certain content was prioritized, how context influenced encoding, and which features contributed to semantic understanding. This transparency is particularly crucial in healthcare, financial services, and critical infrastructure applications where algorithmic decisions must be auditable and justifiable. Future research should develop XAI methods specifically tailored for semantic communication systems that operate within O-RAN's real-time constraints while providing meaningful explanations to network operators and regulators.

5.4.4. Hardware-algorithm co-design and experimental validation

Hardware-algorithm co-design represents a critical research direction for efficient semantic processing. Specialized accelerators for transformer architectures, graph neural networks, and neuromorphic computing must be optimized for O-RAN deployment constraints, balancing computational performance with energy efficiency. Research should quantify trade-offs between different AI architectures (e.g., transformers vs. GNNs) on various hardware platforms and explore integration with emerging technologies like in-memory computing for latency-sensitive applications.

Experimental validation requires advanced testbeds combining real O-RAN hardware with sophisticated simulation capabilities to evaluate semantic protocols under realistic conditions. These platforms should support performance assessment across temporal and spatial scales, including adverse conditions like network congestion and attacks. Large-scale field trials are essential to quantify real-world benefits and identify implementation challenges not apparent in laboratory environments. Prototype deployments in vertical domains such as smart factories, connected vehicles, and telemedicine will provide crucial validation of semantic communication benefits under practical constraints [55,102,103].

5.4.5. Integrative architectures and standardization pathways

The development of unified semantic-AI architectures represents the most promising avenue for addressing current fragmentation. Future research should focus on creating comprehensive frameworks that seamlessly integrate semantic processing capabilities, adaptive learning mechanisms, and AI-driven decision-making within O-RAN's disaggregated architecture. These integrative frameworks must address the fundamental challenge of coordinating semantic understanding across multiple network layers and vendor implementations while maintaining the flexibility and openness that define O-RAN principles [57,104,105].

Standardization pathways must be established through collaboration between academic researchers, industry consortia, and standards organizations. This includes proposing concrete extensions to O-RAN specifications for semantic capabilities, developing interoperability testing frameworks, and establishing certification processes for semantic communication components. The path toward semantic-aware O-RAN/6G networks requires coordinated efforts across these research directions, addressing technical challenges while advancing theoretical foundations and practical deployment frameworks. Success in this endeavor will unlock the transformative potential of semantic communication, enabling networks that intelligently preserve and prioritize meaning to meet the stringent demands of next-generation applications.

6. Conclusion

This survey reveals a critical mismatch between the promising capabilities of semantic communication and the practical requirements for O-RAN deployment. While semantic protocols demonstrate substantial bandwidth improvements across surveyed studies, our analysis identifies fundamental gaps in current O-RAN specifications that prevent widespread adoption. The E2, A1, and O1 interfaces lack semantic information exchange capabilities, semantic KPIs remain unstandardized, and no framework exists for semantic protocol negotiation across multi-vendor environments representing the most significant barrier to commercial deployment. Our comprehensive analysis exposes three critical deployment obstacles. First, computational complexity creates a fundamental bottleneck, with semantic processing demanding significantly higher energy consumption than conventional protocols while transformer-based approaches exhibit quadratic scaling that challenges real-time edge deployment. Second, interoperability across O-RAN's disaggregated architecture remains unsolved, as vendors implement incompatible semantic models and knowledge representations. Third, security vulnerabilities unique to semantic communication-including context manipulation and semantic poisoning attacks-lack adequate defensive frameworks, with the majority of surveyed works failing to address these critical threats. The path forward requires prioritizing lightweight federated semantic learning architectures that can operate within O-RAN's distributed computational constraints while maintaining standardized interfaces. The most promising research direction involves developing hybrid edge-cloud semantic processing frameworks that decompose semantic tasks across the network hierarchy, enabling progressive deployment without requiring wholesale infrastructure replacement. Standardization efforts must focus on extending current O-RAN specifications with semantic-aware interface definitions and establishing measurable semantic quality metrics before widespread adoption becomes feasible.

CRediT authorship contribution statement

Abdellah Tahenni: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Messaoud Ahmed Ouameur:** Validation, Supervision, Conceptualization; **Miloud Bagaa:** Project administration, Methodology; **Daniel Massicotte:** Visualization, Validation, Resources, Project administration; **Sifeddine Salmi:** Writing – review & editing, Validation, Software, Resources; **Felipe Augusto Pereira de Figueiredo:** Visualization, Validation, Software, Resources; **Adlen Ksentini:** Formal analysis, Data curation, Conceptualization.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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